

1. What is the World Wide Web (WWW)?

The **WWW** is a hypertext-based system that provides a uniform and user-friendly interface for accessing resources on the Internet.

It is an **information space** in which items of interest (called **resources**) are identified by global identifiers known as **Uniform Resource Identifiers (URI)**.

2. Architecture of WWW

The architecture of the WWW is **two-tiered**. It consists of a **Client** and a **Server**. The client (web browser) requests a web page; the page is retrieved from the server and returned to the client.



Figure: Two-tier Client–Server architecture of the WWW

Note: When web pages interact with a database, the architecture becomes **three-tiered** (Client → Server → Database).

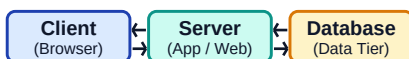


Figure: Three-tier architecture (Client – Server – Database)

3. Three Key Standards of WWW Architecture

The architecture depends on three fundamental standards:

- **HTML** (HyperText Markup Language) — Used for encoding the content of documents so that browsers can display them correctly.
- **URL** (Uniform Resource Locator) — Used for naming remote information objects in a global name space. A URL is a specific type of URI.
- **HTTP** (HyperText Transfer Protocol) — Used for staging the transfer of data between client and server. It defines how the browser and web server communicate.

4. Working of WWW (Client–Server Approach)

Whenever a user wants to retrieve a web page, the WWW works as follows:

- **(i)** User enters the **URL** of the web page in the address bar of the web browser. The browser then requests the **Domain Name Server (DNS)** for the IP address corresponding to the domain (e.g. www.google.com).
- **(ii)** After receiving the IP address, the browser sends a request for the web page to the Internet using the **HTTP protocol**, which specifies the way the browser and web server communicate.
- **(iii)** Internet routers send the request to the intended web server.
- **(iv)** The web server receives the request using the HTTP protocol.
- **(v)** The server examines its hard disk / memory. If the requested file is found, it returns the file to the web browser and closes the HTTP connection.
- **(vi)** The web browser interprets the received file and displays the content of the web page in the browser window.



Figure: Simplified request–response flow of the WWW

5. HTTP – HyperText Transfer Protocol

HTTP is an Internet standard (a set of rules) that allows the exchange of information on the WWW.

HTTP defines **how messages are formatted and transmitted** and **what actions** web servers and browsers should take in response to various commands.

Example: When a user enters a URL, the browser actually sends an HTTP command to the web server directing it to fetch and transmit the requested web page.

6. Evolution of the Web

The web has evolved from a simple content-delivery system (Web 1.0) to a highly interactive, collaborative platform (Web 2.0). The transition from Web 1.0 to Web 2.0 is considered beneficial because it enables interaction and collaboration among users and supports user-generated content in virtual communities.

7. Web 1.0

Web 1.0 is the **first stage** of World Wide Web evolution. According to Tim Berners-Lee, **Web 1.0 is a read-only web**.

Early web allowed users mainly to search and read content that had been published. There was very little user interaction or content generation. In Web 1.0, advertisement while surfing the Internet was restricted in the original vision.

Key characteristics of Web 1.0:

- Only a few content creators; the huge majority of users were consumers of content.
- Personal web pages were common, consisting of **static pages** hosted on ISP-run servers or free web-hosting services.
- Web hosting, publishing and advertising were typical activities.
- Frames and tables were widely used to position and align content on a page.
- Content is served from the server's file system.
- Pages were often built using server-side technologies / Common Gateway Interface (CGI).
- It functioned largely as a content-delivery network enabling the exchange of information.

Design essentials of a Web 1.0 site (from notes):

- Static pages
- Content served from the server file system
- Pages built using server-side / Common Gateway technologies (with possible client-side scripts)

8. Web 2.0

Web 2.0 refers to WWW websites that emphasise **user-generated content**, **usability** and **interoperability**. It is also called the **Participative Social Web**.

Web 2.0 does **not** refer to a change in any technical specification; rather it describes the cumulative changes in the way software developers and end-users design and use the web.

New browser technologies (including JavaScript frameworks) are heavily used in Web 2.0 development.

Five major features of Web 2.0:

- **Free sharing of information** – permits users to receive and classify information collectively.
- **Dynamic content** that is responsive to human-user input.
- **Information flows** between site owners and site users through collaboration and online commenting.
- **Wider user base** – web access moves beyond the traditional Internet user base to a much broader variety of users.
- **Developed APIs** that allow software to search and treat the web as a software application platform.

Common uses / applications of Web 2.0:

- Podcasting · Blogging · Tagging · Social Networking · Social Media · Social Bookmarking · related collaborative services

Quick Comparison – Web 1.0 vs Web 2.0

Aspect	Web 1.0	Web 2.0
Nature	Read-only (Berners-Lee)	Read-write / Participative
Content	Mostly static pages	Dynamic, user-generated
Users	Consumers of content	Creators + collaborators
Interaction	Very limited	High – comments, sharing, APIs
Focus	Publishing & delivery	Collaboration & community

Table: Core differences emphasised in the lecture notes

9. Quick Revision Checklist

- WWW = hypertext system providing a uniform interface to Internet resources identified by URIs.
- Architecture is fundamentally two-tier (Client + Server). Becomes three-tier when a database is added.
- Three key standards: HTML (content), URL (naming), HTTP (transfer).
- Working steps: Enter URL → DNS → HTTP request → Router → Server returns file → Browser renders.
- HTTP = rules for formatting messages and defining actions of browsers and servers.
- Web 1.0 = first stage, read-only (Berners-Lee), static pages, few creators, many consumers.

- Web 2.0 = Participative Social Web – user-generated content, dynamic, APIs, collaboration.
- Five Web 2.0 features: free sharing, dynamic content, owner–user info flow, wider user base, APIs.
- Typical Web 2.0 uses: blogging, podcasting, tagging, social networking, social media, bookmarking.